

Zenith Aero: Grand Voyager — Professional Fabrication & Investment Dossier

This document serves as both a technical fabrication manual and a high-fidelity design proposal for the **Zenith Aero: Grand Voyager (305cm LWB)**. It integrates sustainable bamboo engineering with high-performance carbon-fiber reinforcement and artisan rattan suspension.

1. Design Concept & Hero Visual

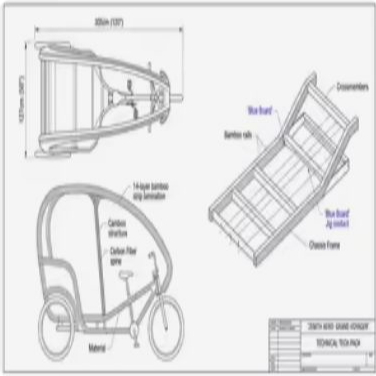
The Grand Voyager is the flagship of the Zenith Aero series, specifically designed for luxury resort mobility and high-end eco-tourism. Its elongated wheelbase (305cm) provides industry-leading passenger legroom and a "floating" cabin aesthetic that distinguishes it from traditional steel-frame pedicabs.

The original high-fidelity artistic rendition of the Grand Voyager aesthetic:

2. Chassis Engineering: The Grand Voyager (LWB)

The 305cm chassis is engineered using a 14-layer artisan strip-lamination process. An internal Carbon Fiber "Spine" (3K UD tape) is embedded within the bamboo layers to provide the rigidity required for the extended span and 250kg payload.

Technical orthographic views and primary dimensions:



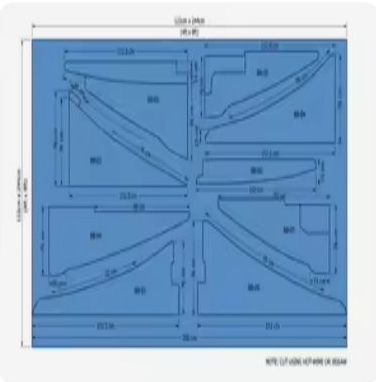
Technical Specifications

- **Total Length:** 305cm (120")
- **Chassis Architecture:** Hybrid Bamboo-Carbon "Artisan Monocoque"
- **Seating:** Suspended Woven Rattan Bench (Coolite Grade)
- **Payload Capacity:** 250kg (Driver + 2 Adult Passengers)

3. Fabrication Methodology: "Blue Board" Molding

To achieve the complex curves of the Voyager while maintaining the strength of continuous bamboo fibers, we utilize high-density XPS "Blue Board" jigs. This method ensures symmetry and structural precision without the high overhead of CNC tooling.

Nested cutting templates for standard 4ft x 8ft XPS foam sheets:



Fabrication Workflow

1. **Molding:** Nested BB-01 and BB-02 templates form the primary longitudinal rails .
2. **Compression:** Layers are compressed onto the templates using industrial heat-shrink tubing and localized heat guns for a 360-degree uniform clamp.
3. **Curing:** Utilizing a slow-cure (24-36 hour) bio-epoxy system (Super Sap CCR/CCS) to allow for the meticulous artisan layup of 14 continuous layers [Swellcomposites](#).

4. Artisan Joint Sculpting & Finish

The "Overbuild-and-Grind" strategy allows for maximum safety at critical stress points while maintaining an organic, aerodynamic silhouette.

Manufacturing workflow for joint sculpting and finish:



Sculpting Procedure

1. **Initial Assembly:** Joints are overbuilt with structural epoxy and carbon fiber wraps .
2. **Precision Grinding:** A Dremel with tungsten carbide burrs is used to sculpt the "Aero-Organic" transitions [Mtbr](#).
3. **Refining:** Final "teeth" for re-coating are created using 80-grit abrasives and the "shoe-shine" hand-sanding method [Bona](#).

5. Local Supply Chain & Investment Value (Vancouver, BC)

The Zenith Aero project is uniquely positioned to utilize a local, sustainable supply chain in the Greater Vancouver Area:

Component	Partner	Value Proposition
Structural Bamboo	Coolite Bamboo	Premium heat-treated poles and art isan weaving Coolitebamboo
Composites & XPS	Industrial Plastics & Paints	High-density foam and 3K Carbon Fi ber supplies Goindustrial
Bio-Resins	Swell Composites	#1 Canadian distributor of eco-frien dly Entropy Resins Swellcomposites

Project Viability

- **Sustainability:** 40% weight reduction over steel frames using renewable bamboo.
- **Innovation:** First-of-its-kind bamboo-carbon-rattan hybrid suspension system.
- **Scalability:** Artisan hand-build method minimizes initial capital expenditure (CAPEX) for the prototype phase.

 Want more tailored results? [Add preference](#)